

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

**Larutan Argentum**
**Silver solution 1000 ppm in 1M nitric acid**

J/8064/15, J/8064/05, J/8064/08

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                        |                     |
|----------------------------------------|---------------------|
| Substances/mixtures corrosive to metal | Category 1 (H290)   |
| Skin Corrosion/Irritation              | Category 1 (H314) B |
| Serious Eye Damage/Eye Irritation      | Category 1 (H318)   |
| Acute aquatic toxicity                 | Category 1 (H400)   |
| Chronic aquatic toxicity               | Category 1 (H410)   |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H410 - Very toxic to aquatic life with long lasting effects

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P363 - Wash contaminated clothing before reuse  
P390 - Absorb spillage to prevent material damage

### Storage

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant inliner

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors  
Toxic to terrestrial vertebrates

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                   | CAS No    | Weight % |
|-----------------------------|-----------|----------|
| Nitric acid ...% [C ≤ 70 %] | 7697-37-2 | 6-7      |
| Water                       | 7732-18-5 | >93      |
| Silver nitrate              | 7761-88-8 | 0.2      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

#### Ingestion

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an

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unconscious person. Call a physician immediately.

## Inhalation

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

## Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

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## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store in metal containers.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                   | Malaysia | ACGIH TLV                   | OSHA PEL                                                                                                                                                              |
|-----------------------------|----------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] |          | TWA: 2 ppm<br>STEL: 4 ppm   | (Vacated) TWA: 2 ppm<br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 4 ppm<br>(Vacated) STEL: 10 mg/m <sup>3</sup><br>TWA: 2 ppm<br>TWA: 5 mg/m <sup>3</sup> |
| Silver nitrate              |          | TWA: 0.01 mg/m <sup>3</sup> | (Vacated) TWA: 0.01 mg/m <sup>3</sup>                                                                                                                                 |

| Component                   | European Union                                             | The United Kingdom                                          | Germany                                                                        |
|-----------------------------|------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------|
| Nitric acid ...% [C ≤ 70 %] | STEL: 1 ppm (15min)<br>STEL: 2.6 mg/m <sup>3</sup> (15min) | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min    | TWA: 1 ppm (8 Stunden). AGW -<br>TWA: 2.6 mg/m <sup>3</sup> (8 Stunden). AGW - |
| Silver nitrate              | TWA: 0.01 mg/m <sup>3</sup> (8hr)                          | TWA: 0.01 mg/m <sup>3</sup> STEL:<br>0.03 mg/m <sup>3</sup> | VME: 0.01 mg/m <sup>3</sup>                                                    |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

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**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

**Recommended Filter type:** Particulates filter conforming to EN 143 or Acid gases filter Type E Yellow conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                          |                                          |
|-------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                   | Colorless                |                                          |
| <b>Physical State</b>               | Liquid                   |                                          |
| <b>Odor</b>                         | No information available |                                          |
| <b>Odor Threshold</b>               | No data available        |                                          |
| <b>pH</b>                           | < 1 (estimated)          |                                          |
| <b>Melting Point/Range</b>          | No data available        |                                          |
| <b>Softening Point</b>              | No data available        |                                          |
| <b>Boiling Point/Range</b>          | No information available |                                          |
| <b>Flash Point</b>                  | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | No data available        |                                          |
| <b>Flammability (solid,gas)</b>     | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>             | No data available        |                                          |
| <b>Vapor Pressure</b>               | No data available        |                                          |
| <b>Vapor Density</b>                | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>   | No data available        |                                          |
| <b>Bulk Density</b>                 | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>             | Miscible                 |                                          |
| <b>Solubility in other solvents</b> | No information available |                                          |

### Partition Coefficient (n-octanol/water)

| Component                   | log Pow |
|-----------------------------|---------|
| Nitric acid ...% [C ≤ 70 %] | -2.3    |
| Silver nitrate              | 0.19    |

|                                  |                          |
|----------------------------------|--------------------------|
| <b>Autoignition Temperature</b>  | No data available        |
| <b>Decomposition Temperature</b> | No data available        |
| <b>Viscosity</b>                 | No data available        |
| <b>Explosive Properties</b>      | No information available |
| <b>Oxidizing Properties</b>      | No information available |

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

#### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat.

### Incompatible Materials

Strong bases. Strong reducing agents. Metals.

### Hazardous Decomposition Products

Nitrogen oxides (NOx).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

#### **(a) acute toxicity;**

##### **Oral**

Based on available data, the classification criteria are not met

##### **Dermal**

Based on available data, the classification criteria are not met

##### **Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component                   | LD50 Oral            | LD50 Dermal               | LC50 Inhalation              |
|-----------------------------|----------------------|---------------------------|------------------------------|
| Nitric acid ...% [C ≤ 70 %] | -                    | -                         | LC50 = 2500 ppm. (Rat) 1h    |
| Water                       | -                    | -                         | -                            |
| Silver nitrate              | > 2000 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | LC50 > 750 µg/m³ ( Rat ) 4 h |

| Component                   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-----------------------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid ...% [C ≤ 70 %] | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

ECHA (RAC) - Committee for Risk Assessment - European CHemicals Agency

ATE - Acute Toxicity Estimate

#### **(b) skin corrosion/irritation;**

Category 1 B

#### **(c) serious eye damage/irritation;**

Category 1

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**(d) respiratory or skin sensitization;**

Respiratory No data available  
Skin No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

Target Organs None known.

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component      | Freshwater Fish                      | Water Flea            | Freshwater Algae | Microtox                                                                                                                                                                                                                           |
|----------------|--------------------------------------|-----------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silver nitrate | Leuciscus idus: LC50: 0.029 mg/L/96h | EC50: 0.0006 mg/L/48h |                  | Photobacterium phosphoreum: EC50: 0.038 mg/L/24h<br>Photobacterium phosphoreum: EC50: 0.395 mg/L/15min<br>Photobacterium phosphoreum: EC50: 0.44 mg/L/30 min as Ag++<br>Photobacterium phosphoreum: EC50: 0.86 mg/L/15 min as Ag++ |

**Persistence and degradability**

**Persistence** Soluble in water, Persistence is unlikely, based on information available, Miscible with water.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

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|                                  |                |                                      |
|----------------------------------|----------------|--------------------------------------|
| <b>Bioaccumulative potential</b> |                | Bioaccumulation is unlikely          |
| <b>Component</b>                 | <b>log Pow</b> | <b>Bioconcentration factor (BCF)</b> |
| Nitric acid ...% [C ≤ 70 %]      | -2.3           | No data available                    |
| Silver nitrate                   | 0.19           | No data available                    |

**Mobility in soil** The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations Should not be released into the environment

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Large amounts will affect pH and harm aquatic organisms Solutions with low pH-value must be neutralized before discharge Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
UN-No UN3264  
Hazard Class 8  
Packing Group II  
Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains NITRIC ACID)

**Road and Rail Transport**  
UN-No UN3264  
Hazard Class 8  
Packing Group II  
Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains NITRIC ACID)

**IATA**  
UN-No UN3264  
Hazard Class 8  
Packing Group II  
Proper Shipping Name CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains NITRIC ACID)

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

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## Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed

| Component                   | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Nitric acid ...% [C ≤ 70 %] | 231-714-2 | X    | X   | X     | X    | X    | X     | X    | KE-25911 |
| Water                       | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Silver nitrate              | 231-853-9 | X    | X   | X     | X    | X    | X     | X    | KE-31281 |

| Component                   | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Major<br>Accident Notification | Seveso III Directive<br>(2012/18/EC) - Qualifying<br>Quantities for Safety<br>Report Requirements | Rotterdam Convention<br>(PIC) | Basel Convention<br>(Hazardous Waste) |
|-----------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Nitric acid ...% [C ≤ 70 %] |                                                                                                    |                                                                                                   |                               | Annex I - Y34                         |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

Revision Date

23-Mar-2025

Revision Summary

SDS sections updated, 3, 14.

**In accordance with local and national regulations: Occupational Safety and Health**

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## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**